

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

SOFTWARE ENGINEER – HCL Technologies | Chennai, India

09/2021 – 01/2025 (3.5 Years)

- Developed and implemented **translational medicine strategies** and **biomarker discovery protocols** using **Python data pipelines**, **R statistical analysis**, and **Power BI dashboards** to interpret preclinical and clinical study data, supporting drug development program objectives with quantitative rigor across 8+ therapeutic candidates at HCL's enterprise healthcare clients.
- Designed and validated **biomarker assay development programs** and **investigative translational strategies** by coordinating with **Contract Research Organizations (CROs)**, managing external partner timelines and quality standards, and ensuring **regulatory expectations** alignment for IND submissions and pre-IND meeting documentation across 12+ CRO engagements.
- Prepared and reviewed **IND supporting documentation**, **non-clinical sections of regulatory documents**, protocols, and **Investigator's Brochures** in collaboration with clinical development and regulatory teams, incorporating **biomarker study results** and **mechanism of action data** to support asset readiness for clinical development phases.
- Led **multifunctional translational science programs** bridging **drug discovery** and **clinical programs** by synthesizing preclinical data, clinical biomarker findings, and pharmacology insights using **SQL data warehousing**, **DAX measures**, and **Tableau visualizations** to drive key program decision-making with 95% stakeholder alignment.
- Managed **experimental program execution** and **bioanalytical program coordination** with internal discovery, pharmacology, and clinical teams using **Jira ticketing**, **change-control protocols**, and **Quality Management & Assurance workflows** to ensure qualified vendor performance and departmental standardizations across preclinical and clinical asset pipelines.
- Presented scientific findings and translational strategy recommendations internally and externally at regulatory briefing meetings and scientific conferences, communicating complex **biomarker validation data**, **assay performance metrics**, and **clinical trial support outcomes** to drive organizational decision-making and advance therapeutic programs with executive alignment.

SKILLS

Translational Medicine & Drug Development – biomarker discovery, biomarker validation, assay development, translational strategy design, preclinical-to-clinical bridging, mechanism of action assessment, IND supporting documentation, Investigator's Brochures, pre-IND meeting preparation, regulatory strategy.

Biomarker & Bioanalytical Science – biomarker implementation, biomarker measures for animal and patient studies, drug response assessment, bioanalytical program coordination, assay performance evaluation, clinical trial biomarker support.

Regulatory & Compliance – IND submissions, regulatory documentation preparation, regulatory expectations alignment, pre-IND meetings with regulatory authorities, non-clinical sections of regulatory dossiers, briefing meeting packages, submission dossiers, Quality Management & Assurance protocols.

Program Management & External Partnerships – Contract Research Organization (CRO) management, external partner coordination, experimental program planning and execution, multifunctional team leadership, change-control management, vendor quality assurance, departmental standardization.

Data Analysis & Scientific Communication – preclinical data interpretation, clinical biomarker data synthesis, Power BI dashboards, Tableau visualizations, R statistical analysis, Python data pipelines, SQL data warehousing, DAX measures, scientific presentation, regulatory communication.

Technical Stack – Python, R, SQL (PostgreSQL, MySQL, Oracle), Power BI, Tableau, Jira, Git, Linux, AWS (EC2, S3, Lambda), Docker, pandas, NumPy, scikit-learn, statistical analysis, hypothesis testing, data modelling.

Healthcare Domain Expertise – plasma-derived medicines, transfusion medicine, drug development workflows, clinical trial support, asset readiness assessment, therapeutic program advancement, healthcare company operations.

EDUCATION AND TRAINING

MASTER OF SCIENCE in Data Analytics – National College of Ireland, Dublin	02/2026
BACHELOR OF ENGINEERING in Computer Science – SNS College of Technology, Coimbatore, India	01/2021

PROJECTS

Biomarker Discovery & Clinical Trial Support Platform (HCL Technologies)

- Architected a comprehensive **translational medicine data platform** integrating preclinical biomarker assay results, animal study pharmacology data, and clinical trial biomarker findings using **Python ETL pipelines** and **SQL data warehousing** to consolidate 45+ data sources from internal discovery teams and 8 external CROs, enabling unified biomarker validation workflows.
- Designed and deployed **interactive Power BI dashboards** with **DAX measures** for real-time biomarker performance tracking, mechanism of action visualization, and assay development progress monitoring, reducing regulatory documentation preparation time by **40%** and enabling data-driven decision-making across 12+ therapeutic programs.
- Implemented **biomarker validation protocols** and **assay development governance** frameworks using **Jira change-control workflows** and **Quality Management & Assurance checklists** to ensure CRO compliance with regulatory expectations, achieving **100% audit-ready documentation** for 6 IND submissions within 18-month period.
- Developed statistical analysis and hypothesis testing modules in **R** to interpret preclinical study biomarker data, calculate drug response indicators, and validate biomarker measures for clinical asset readiness, supporting **3 pre-IND regulatory meetings** with quantitative evidence packages.
- Coordinated multifunctional translational science programs bridging discovery, pharmacology, and clinical teams through **Confluence documentation** and **AWS-hosted study tracking dashboards**, reducing cross-functional communication cycles by **35%** and accelerating asset transition to clinical development phases.

Regulatory Documentation & IND Support Automation (HCL Technologies)

- Streamlined **IND supporting documentation preparation** and **Investigator's Brochure compilation** by building **Python automation scripts** to extract, validate, and standardize non-clinical biomarker data, pharmacology findings, and assay performance metrics from heterogeneous enterprise systems, reducing manual documentation time by **50%**.
- Collaborated with regulatory and clinical teams to define **translational strategy** templates and **pre-IND meeting briefing packages**, incorporating synthesized biomarker validation results and mechanism of action summaries into submission dossiers that achieved **zero regulatory comments** on scientific rigor for 4 consecutive IND submissions.

- Established **departmental standardization** protocols for bioanalytical program documentation and CRO quality requirements using **Jira ticketing** and **Git version control** for regulatory document change management, ensuring **99%** compliance with Quality Management & Assurance standards across external vendor submissions.

ACHIEVEMENTS & CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California, Davis** (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)